

10. Collusive Oligopoly

One way of avoiding the uncertainty arising from oligopolistic interdependence is to enter into collusive agreements. There are two main types of collusion, cartels and price leadership. Both forms generally imply tacit (secret) agreements, since open collusive action is commonly illegal in most countries at present.

Although direct agreements among the oligopolists are the most obvious examples of collusion, in the modern business world trade associations, professional organisations and similar institutions usually perform many of the activities and achieve in a legal or indirect way the goals of direct collusive agreements. For example, trade associations issue various periodicals with information concerning actual or planned action of members. In this official way firms get the message and act accordingly.

In this chapter we will examine the two formal types of collusion, cartels and price leadership. Both forms have been exhaustively analysed by W. Fellner.¹

I. CARTELS

We saw that, in the absence of collusion, the monopoly solution in the industry (the solution at which the joint industry profit is maximised) can be achieved under the rare conditions that (a) each firm knows the monopoly price, that is, has a correct knowledge of the market demand and of the costs of all firms, (b) that each firm recognises its interdependence with the others in the industry, (c) all firms have identical costs and identical demands. (Actually condition (c) implies condition (a).)

We will examine two typical forms of cartels: (a) cartels aiming at joint-profit maximisation, that is, maximisation of the industry profit, and (b) cartels aiming at the sharing of the market.

A. CARTELS AIMING AT JOINT-PROFIT MAXIMISATION

Cartels imply direct (although secret) agreements among the competing oligopolist with the aim of reducing the uncertainty arising from their mutual interdependence. In this particular case the aim of the cartel is the maximisation of the industry (joint) profit. The situation is identical with that of a multiplant monopolist who seeks the maximisation of his profit. We concentrate on a homogeneous or pure oligopoly, that is, an oligopoly where all firms produce a homogeneous product. The case of differentiated oligopoly will be examined in a separate section.

The firms appoint a central agency, to which they delegate the authority to decide not only the total quantity and the price at which it must be sold so as to attain maximum

¹ W. Fellner, *Competition Among the Few* (New York, Knopf, 1949).

profits, but also the allocation of production among the members of the cartel, and the distribution of the maximum joint profit among the participating members. The authority of the central cartel agency is complete. Clearly the central agency will have access to the cost figures of the individual firms, and for the purposes of the present theory we unrealistically suppose that it will calculate the market-demand curve and the corresponding MR curve. From the horizontal summation of the MC curves of individual firms the market MC curve is derived. The central agency, acting as a multi-plant monopolist, will set the price defined by the intersection of the industry MR and MC curves. For simplicity assume that there are only two firms in the cartel. Their cost structure is shown in figures 10.1 and 10.2. This is implied by the profit-maximisation goal of the cartel: each level of industry output should be produced at the least possible cost. Thus if we add the outputs of A and B that can be produced at the same MC, clearly the resulting total output is the output that can be produced at this common lowest cost. Given the market demand DD (in figure 10.3) the monopoly solution,

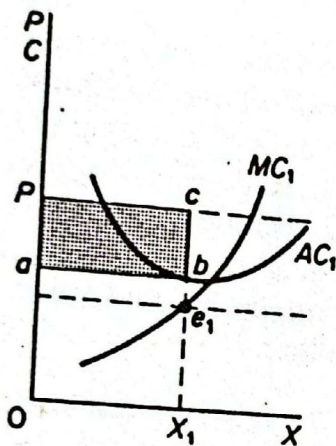


Figure 10.1

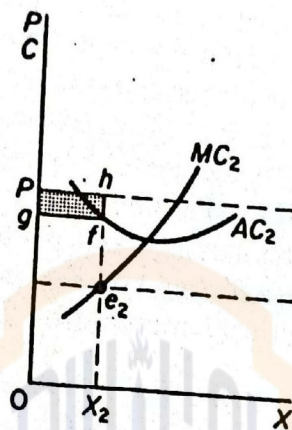


Figure 10.2

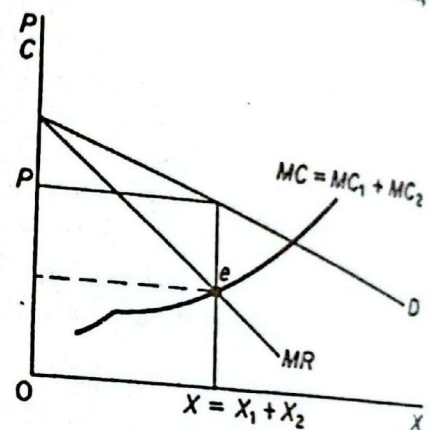


Figure 10.3

which maximises joint profits, is determined by the intersection of MC and MR (point e in figure 10.3). The total output is X and it will be sold at price P. Now the central agency allocates the production among firm A and firm B as a monopolist would do, that is, by equating the MR to the individual MCs. Thus firm A will produce X₁ and firm B will produce X₂. Note that the firm with the lower costs produces a larger amount of output. However, this does not mean that A will also take the larger share of the attained joint profit. The total industry profit is the sum of the profits from the output of the two firms, denoted by the shaded areas of figures 10.1 and 10.2. The distribution of profits is decided by the central agency of the cartel.

The mathematical presentation of the cartel model which aims at joint profit maximisation is identical to that of the multiplant monopolist. Thus:

$$\text{Maximise } \Pi = \Pi_1 + \Pi_2$$

given

$$\begin{aligned} P &= f(X) = f(X_1 + X_2) \\ C_1 &= f_1(X_1) \\ C_2 &= f_2(X_2) \end{aligned}$$

We have

$$\begin{aligned} \Pi_1 &= R_1 - C_1 \\ \Pi_2 &= R_2 - C_2 \end{aligned}$$

Therefore

$$\Pi = R_1 + R_2 - C_1 - C_2 = R - C_1 - C_2$$

and the market marginal revenue is

$$\frac{\partial R}{\partial X} = \frac{\partial R}{\partial X_1} = \frac{\partial R}{\partial X_2}$$

that is, each additional unit will bring the same MR , irrespective of the plant in which it is produced, since all units of X are sold at the same price P .

The first-order condition for maximisation of the joint profit Π requires the allocation of output in such a way that the MC of each firm is the same:

$$\frac{\partial \Pi}{\partial X_1} = \frac{\partial R}{\partial X} - \frac{\partial C_1}{\partial X_1} = 0 \rightarrow \frac{\partial R}{\partial X} = \frac{\partial C_1}{\partial X_1}$$

$$\frac{\partial \Pi}{\partial X_2} = \frac{\partial R}{\partial X} - \frac{\partial C_2}{\partial X_2} = 0 \rightarrow \frac{\partial R}{\partial X} = \frac{\partial C_2}{\partial X_2}$$

or

$$MR = MC_1 = MC_2$$

The second-order condition for maximisation of joint Π requires

$$\frac{\partial^2 R}{\partial X^2} < \frac{\partial^2 C_1}{\partial X_1^2} \quad \text{and} \quad \frac{\partial^2 R}{\partial X^2} < \frac{\partial^2 C_2}{\partial X_2^2}$$

that is, the MC of each firm must be increasing faster than the (common) MR of the output of the cartel as a whole.
These are the same equilibrium conditions as the ones we derived in Chapter 6 for the multiplant monopolist.

Numerical Example

Assume that the market demand is

$$P = 100 - 0.5(X) = 100 - 0.5(X_1 + X_2) \quad (10.1)$$

and that the two colluding firms have costs given by

$$C_1 = 5X_1, \quad \text{and} \quad C_2 = 0.5X_2^2$$

The central agency of the cartel aims at the maximisation of total profit

$$\Pi = \Pi_1 + \Pi_2$$

where

$$\Pi_1 = R_1 - C_1 \quad \text{and} \quad \Pi_2 = R_2 - C_2$$

Thus

$$\begin{aligned} \Pi &= (R_1 + R_2) - C_1 - C_2 \\ \Pi &= P(X_1 + X_2) - C_1 - C_2 \\ \Pi &= [100 - 0.5(X_1 + X_2)](X_1 + X_2) - 5X_1 - 0.5X_2^2 \\ \Pi &= 95X_1 - 100X_2 - 0.5X_1^2 - X_2^2 - X_1X_2 \end{aligned} \quad (10.2)$$

Setting the partial derivatives equal to zero we obtain

$$\frac{\partial \Pi}{\partial X_1} = 85 - X_1 - X_2 = 0$$

$$\frac{\partial \Pi}{\partial X_2} = 100 - X_1 - 2X_2 = 0$$

Solving for X_1 and X_2 we find

$$X_1 = 90 \quad \text{and} \quad X_2 = 5$$

The price is found by substituting in (10.1)

$$P = 100 - 0.5(X_1 + X_2) = 52.5$$

The joint profit may be obtained by substituting in (10.2)

$$\Pi = 95(90) - 100(5) - 0.5(90)^2 - (5)^2 - (90)(5) = 4525$$

Although theoretically the monopoly solution is easy to derive, in practice cartels rarely achieve maximum joint profits. There are several reasons why industry profits cannot be maximised, even with direct cartel collusion.¹

It should be obvious that equilibrium with joint profit maximisation will be easier to attain and will be in general stable if firms have identical costs and identical demands, conditions that are rarely met in practice. (However, even with identical costs cartels may still be unstable. See below.) Even in these conditions many factors may mitigate against achievement of the joint profit-maximisation goal of the cartel. The main reasons why industry profits may not be maximised may be summarised as follows.

First. Mistakes in the estimation of market demand. Usually the elasticity of market demand is underestimated. Each firm believes that the e of its own demand curve is high due to the existence of perfect (or almost perfect) substitutes made by competitors, while the industry demand is much less elastic. Clearly, mistakes in the estimation of the market demand lead to mistakes in the derivation of the MR and hence to a price which is higher than the monopoly price.

Second. Mistakes in the estimation of MC . The estimation of the market MC , from the summation of individual costs, may involve mistakes, due to incomplete knowledge of the individual MC curves at all levels of output. That is, even if a cartel estimated marginal costs, which is highly improbable, it would probably get them wrong. Such errors lead to an equilibrium which differs from the monopoly solution. There is a strong incentive for the individual members to present low-cost figures to the central agency, since the allocation of output and of profit shares is determined, among others, by the level of costs.

Third. Slow process of cartel negotiations. Cartel agreements take a long time to negotiate due to the differences in size, costs, and markets of the individual firms. During the negotiations each firm is bargaining in order to attain the greatest advantage from the cartel agreement. Thus, even if at the beginning of the negotiations costs and market demand were correctly estimated, by the time agreement is reached market conditions may have changed, thus rendering the initially defined monopoly price obsolete. Cartel agreements with more than about twenty partners are difficult to reach, and break down easily once reached.

Fourth. 'Stickiness' of the negotiated price. Once the agreement about price is reached, its level tends to remain unchanged over long periods, even if market conditions are

¹ See W. Fellner, *Competition Among the Few* (New York, Knopf, 1949).